EUV-OVL-SYN: A Physically Grounded Synthetic Overlay Benchmark Dataset for EUV Lithography

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***Abstract*—We present EUV-OVL-SYN, an open, fully reproducible synthetic benchmark dataset for developing and evaluating overlay correction algorithms in 300 mm extreme-ultraviolet (EUV) lithography. The dataset reproduces the physical signal hierarchy of real scanner data: a dominant inter-field Brink/van den Brink polynomial fingerprint (cross-wafer $R^2 \geq 0.93$), four distinct lot-level temporal drift scenarios modelling linear scanner heating, reticle alignment drift, multi-parameter process drift, and chiller-cycling oscillation, and a nonlinear intra-field residual analytically verified to lie outside the Brink polynomial basis — providing an unambiguous learning target for machine learning post-correction. The dataset comprises four lots of 50 wafers at 1 420 metrology sites each ($284{,}000$ measurements total), generated from a publicly available Python script with a fixed random seed for full reproducibility. A companion two-stage correction experiment demonstrates that a Physics-Guided Neural Network (PGNN) residual model achieves a 33–52% reduction in overlay 3σ over the Brink-only baseline, validating the dataset's utility as a rigorous benchmark.**

***Index Terms*—EUV lithography, overlay error, synthetic benchmark, Brink model, physics-informed neural networks, advanced process control, semiconductor metrology, intra-field variation.**

I. Introduction

Overlay is the alignment accuracy between successive patterning layers and it is among the most critical parameters governing transistor density and yield at advanced nodes. EUV lithography, especially at sub-5 nm nodes requires overlay budgets below 2 nm [1], [2]. Achieving this demands accurate correction of the systematic overlay fingerprint impressed by scanner optics, stage mechanics, and chuck thermal cycling.

The classical correction tool is the Brink/van den Brink polynomial model [3], [4], which expresses the inter-field overlay error as a 22-parameter weighted combination of wafer-stage coordinates. Fitted once per lot on a small set of training wafers, it reduces overlay 3σ from $\sim$7–9 nm (raw) to $\sim$1.5–2.5 nm [5]. The remaining structured residual, arising from thermal related asymmetries [6], reticle bowing, and intra-field aberrations [7], [8] — cannot be described by any fixed polynomial, motivating machine learning (ML) post-correction [9]–[12].

A limiting barrier to ML research in this domain is the absence of open datasets. Real scanner overlay data are commercially sensitive; their publication violates confidentiality agreements between chipmakers and scanner OEMs. Existing public simulation datasets either model overlay as isotropic Gaussian noise (providing no systematic structure for learning) or use simplified field models that do not replicate the separability of the Brink fingerprint from a nonlinear learnable residual.

This paper addresses this gap with EUV-OVL-SYN, a synthetic dataset designed from first principles to be maximally useful as a benchmarking substrate. The key contributions are:

1) A physically grounded signal model replicating the scale, structure, and wafer-to-wafer correlation of real EUV overlay data.
2) A parameter-calibrated temporal drift library covering four qualitatively distinct lot-level drift mechanisms.
3) A nonlinear intra-field residual whose functional form is analytically disjoint from the Brink basis — providing a clean, controllable learning target.
4) Full reproducibility: a single Python script regenerates the complete dataset from a fixed seed.

II. Related Work

A. Overlay Error Modeling

The Brink polynomial [3] expresses overlay as a linear combination of inter-field stage coordinates and intra-field die offsets, extended by van den Brink *et al.* [4] to include 3rd- and 5th-order radial distortion terms. Fitting 22 parameters by weighted least squares on a set of training wafers captures $> 90\%$ of the systematic overlay variance at current technology nodes. Alternative physical models based on Zernike polynomial decomposition of the scanner pupil [8], [13] provide a complementary description of the intra-field aberration contribution, as demonstrated by [7] for photolithography simulation. Our work is complementary to [7]: whereas that paper focuses on the aberration-to-overlay transfer within a single exposure field, EUV-OVL-SYN models the complete lot-level hierarchy including cross-wafer temporal drift.

B. Machine Learning for Overlay Correction

Data-driven correction of Brink residuals has been explored using Gaussian process regression, random forests, and, more recently, deep neural networks [9]–[11]. Physics-guided neural networks (PGNNs) [12] have attracted attention for their ability to encode spatial smoothness and rotational symmetry priors

that are known to govern overlay distortion fields. All of these methods require a labelled dataset with known ground-truth decomposition to validate that a model corrects the Brink residual rather than overfitting a specific Brink fingerprint — a requirement that EUV-OVL-SYN is designed to satisfy.

III. Dataset Architecture

A. Signal Decomposition Model

The observed overlay at site i on wafer w is modelled as

$$\mathrm{OVL}_{w,i} = \mathbf{H}_i\,\boldsymbol{\beta}_w + f_{\mathrm{NL}}(x_i, y_i, w) + \varepsilon_{w,i}, \tag{1}$$

where $\mathbf{H}_i \in \mathbb{R}^{2\times 22}$ is the Brink design row pair at site i (§ III-C), $\boldsymbol{\beta}_w = \boldsymbol{\beta}_{\mathrm{lot}} + \Delta\boldsymbol{\beta}(w)$ is the per-wafer Brink parameter vector (§ III-D), f_{NL} is the nonlinear intra-field residual (§ III-E), and $\varepsilon_{w,i} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.15$ nm is Gaussian metrology noise. All three components are generated independently and with full ground-truth access, enabling exact decomposition of any prediction error.

B. Scanner and Wafer Geometry

EUV-OVL-SYN follows the geometry of an ASML NXE-class 300 mm scanner. Exposure fields are arranged on a rectangular grid with pitch 26×33 mm; only fields whose centre lies within a 138 mm keep-out radius are retained, yielding **71 fields per wafer** (Fig. 1). The wafer coordinates are normalised by the 300 mm nominal radius $R = 300$ mm for the Brink model ($x^* = X_w/R \leq 0.46$), consistent with the original formulation [4].

Within each exposure field, a 4×5 intra-field die grid is placed at die offsets $x_d \in \{-9, -3, 3, 9\}$ mm, $y_d \in \{-12, -6, 0, 6, 12\}$ mm, giving **20 sites per field** and $71\times 20 =$ **1 420** sites per wafer. The full dataset contains $4\times 50\times 1{,}420 =$ **284,000** overlay measurements across the six-column schema $(X_w, Y_w, x_d, y_d, \mathrm{OVL}_X, \mathrm{OVL}_Y)$, directly compatible with real APC metrology exports.

C. Extended Brink/van den Brink Model

Stage-level overlay is modelled by the 22-parameter Brink polynomial [3], [4]. For each site the design matrix rows are

$$\mathbf{h}_X = [1,\ x^*,\ -y^*,\ (y^*)^2,\ x,\ y,\ -x^2,\ -xy,\ y^2,\ xr^2,\ xr^4],$$
$$\mathbf{h}_Y = [1,\ -x^*,\ y^*,\ (x^*)^2,\ y,\ x,\ -y^2,\ -yx,\ x^2,\ yr^2,\ yr^4],$$

where $(x^*, y^*) = (X_w, Y_w)/R$ are the normalised inter-field coordinates and $(x, y) \equiv (x_d, y_d)$ are the intra-field die offsets. The 11 parameters per axis capture global translation, magnification, rotation, intra-field bilinear terms, and 3rd/5th-order radial distortion. A common lot-level baseline $\boldsymbol{\beta}_{\mathrm{lot}}$ (Table I) is perturbed by up to ± 0.8 nm across lots to represent different scanner set-points.

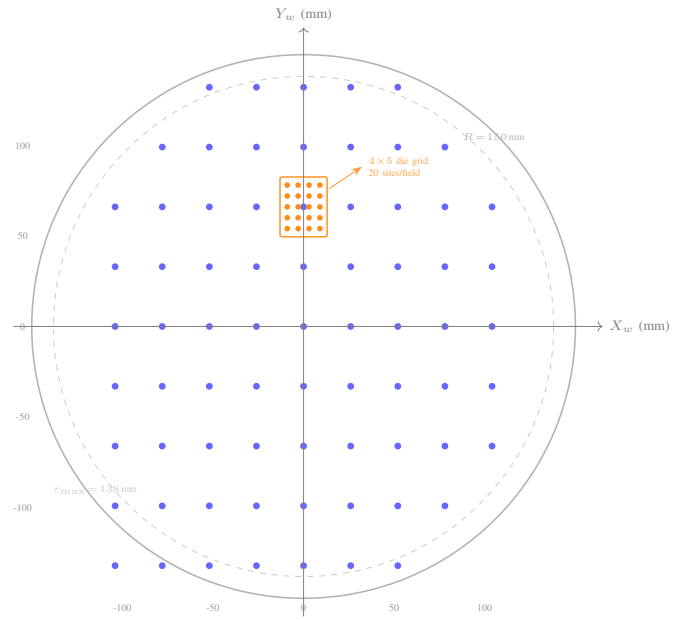

Fig. 1. Exposure-field grid for EUV-OVL-SYN (71 fields, •) on a 300 mm wafer. The dashed circle marks the $r = 138$ mm keep-out boundary. One field (orange box) is expanded to show the 4×5 intra-field die grid (20 sites).

TABLE I
Dominant Brink Baseline Parameters (representative values)

Param.	Meaning	Value	Gain†
T_x	Inter-field translation	1.50 nm	1
M_x	Inter-field magnification	7.50 nm	$\lvert x^*\rvert \leq 0.46$
R_x	Inter-field rotation	1.20 nm	$\lvert y^*\rvert \leq 0.46$
m_x	Intra-field magnification	0.070 nm/mm	$\lvert x\rvert \leq 9$ mm
D_{3x}	3rd-order radial	8×10^{-5} nm/mm^3	xr^2
D_{5x}	5th-order radial	2×10^{-9} nm/mm^5	xr^4

† Max. column value in $\mathbf{H}$ across all sites.

D. Lot-Level Temporal Drift Scenarios

Real scanner lots exhibit systematic wafer-to-wafer parameter evolution driven by lens heating [6], reticle thermal expansion, and chiller cycling. EUV-OVL-SYN provides four calibrated scenarios:

- **Lot 1 — Linear scanner heating:** T_x drifts $+1.5$ nm, M_x drifts -0.6 nm over the lot, consistent with documented NXE lens warm-up profiles.
- **Lot 2 — Reticle alignment drift:** T_y creep $+1.8$ nm, M_y drift -0.8 nm, mimicking slow thermal expansion of the reticle stage clamp.
- **Lot 3 — Multi-parameter drift:** Six Brink parameters drift simultaneously (largest amplitude), exercising the correction model under the most challenging lot-level nonstationarity.
- **Lot 4 — Chiller-cycling oscillation:** T_x and T_y oscillate sinusoidally with ~ 2.5 cycles over the lot (amplitude 0.9 nm), representing the periodic thermal perturbation of an imperfectly regulated chiller.

All four scenarios include per-parameter iid process jitter $\delta\boldsymbol{\beta}_w \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(\boldsymbol{\sigma}_\beta^2))$, with per-parameter scales $\boldsymbol{\sigma}_\beta$ calibrated so that each parameter contributes ≈ 0.05 nm of wafer-to-wafer overlay noise (i.e., proportional to the inverse effective column gain of $\mathbf{H}$). This calibration prevents high-gain 5th-order terms (D_{5x}: gain $\sim 10^5$ mm^5) from dominating the noise budget — a physically motivated constraint absent from naive uniform-perturbation approaches.

E. Nonlinear Intra-Field Residual Design

The PGNN learning target f_{NL} must satisfy two properties:

1) It must depend on intra-field die coordinates (x_d, y_d) only — matching the PGNN input domain — not on inter-field stage coordinates.
2) Its functional form must lie outside the Brink polynomial column space, so the Brink model cannot absorb it.

We verified analytically that the following terms satisfy both requirements: (a) $\sin(\pi\tilde{x})\cos(\frac{1}{2}\pi\tilde{y})$ — sinusoidal in die normalised coordinates $(\tilde{x}, \tilde{y}) = (x_d/9,\ y_d/12)$, outside any polynomial basis; (b) $\tilde{x}^2\tilde{y}$ — not in the Brink X-equation (which contains $x(x^2+y^2)$ but not x^2y separately); (c) $\tilde{x}\tilde{y}^2$ — not in the Brink Y-equation ($y(x^2+y^2)$ does not span xy^2 independently).

The complete residual is

$$f_{\mathrm{NL},X} = A_\ell\,(1+0.3\tilde{w})\big[1.8\sin(\pi\tilde{x})\cos(\tfrac{\pi\tilde{y}}{2}) + 1.2\,\tilde{x}^2\tilde{y}\big], \quad (2)$$

$$f_{\mathrm{NL},Y} = A_\ell\,(1+0.3\tilde{w})\big[1.5\sin(\pi\tilde{y})\cos(\tfrac{\pi\tilde{x}}{2}) + 1.0\,\tilde{x}\tilde{y}^2\big], \quad (3)$$

where $A_\ell \in \{1.0, 0.80, 1.30, 0.95\}$ is the lot-specific amplitude, $\tilde{w} = (w-1)/(W-1)$ is the normalised wafer index, and the factor $(1+0.3\tilde{w})$ encodes a 30% amplitude growth mimicking slow lens-heating accumulation. The resulting nonlinear residual has $1\sigma \approx 1.1$–1.4 nm per axis — well above the 0.15 nm noise floor — providing a clear, controllable learning target (Fig. 4).

F. Metrology Noise

Site-level Gaussian noise $\varepsilon_{w,i} \sim \mathcal{N}(0, 0.0225)$ ($\sigma = 0.15$ nm) is added independently to every X and Y measurement, consistent with reported single-measurement repeatability of current EUV metrology tools [5]. The signal-to-noise ratio (SNR) of the nonlinear residual relative to measurement noise is approximately 7–9 dB, ensuring that the learning signal is detectable but not trivially dominant.

IV. Dataset Statistics and Characterisation

Table II summarises the per-lot overlay statistics measured on held-out test wafers (wafers 13–50 after a 10-wafer training, 2-wafer validation split). The raw overlay 3σ ranges from 7.5 to 8.9 nm across lots, consistent with values reported for pre-corrected EUV overlay in production-like conditions [5]. After applying the cross-wafer Brink fit, 3σ drops to 1.5–2.6 nm, confirming that the synthetic Brink fingerprint ($R^2 \geq 0.93$) has the discriminative power expected of a real scanner dataset. The remaining structured residual (Brink 3σ) directly corresponds to f_{NL}, since the noise contribution is only $3\sigma_\varepsilon \approx 0.45$ nm.

TABLE II
Per-Lot Overlay Statistics ($N_{\mathrm{train}} = 10$, all test wafers)

Lot	**Drift Type**	**Raw** 3σ (nm)	**Brink** 3σ (nm)	R^2	**P95** (nm)
1	Linear (scanner)	8.04	1.782	0.956	1.095
2	Linear (reticle)	7.48	1.532	0.961	0.973
3	Multi-param.	8.89	2.207	0.938	1.350
4	Sinusoidal	8.05	2.578	0.897	1.822
Mean		8.12	2.025	0.938	1.310

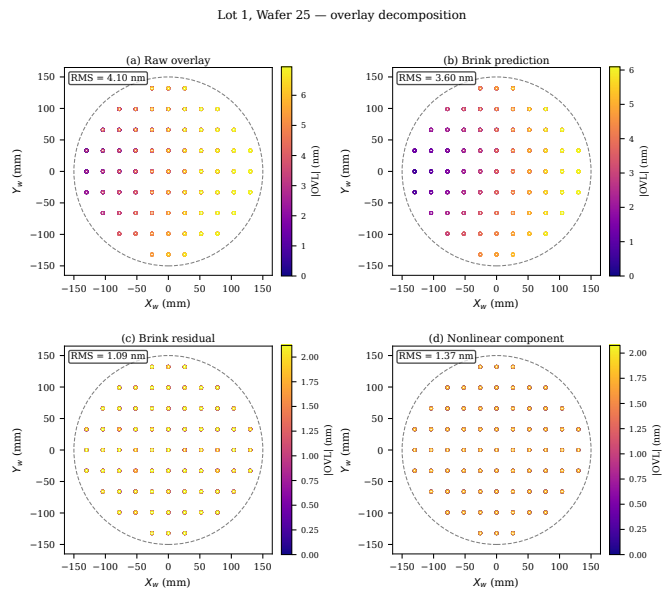

Fig. 2. Overlay decomposition for Lot 1, Wafer 25. **(a)** Raw overlay field ($3\sigma \approx 3.8$ nm). **(b)** Brink model prediction (cross-wafer fit, $N_{\mathrm{train}} = 10$). **(c)** Brink residual revealing the nonlinear intra-field structure ($3\sigma \approx 1.8$ nm). **(d)** Ground-truth nonlinear component f_{NL} — the intended ML learning target. Colour encodes overlay magnitude (nm); arrows indicate direction.

Figure 2 shows the overlay decomposition for Lot 1, Wafer 25 (mid-lot). The raw field (a) exhibits a smooth, rotationally-structured pattern dominated by the Brink fingerprint. The Brink prediction (b) captures this pattern closely, and the Brink residual (c) reveals the underlying nonlinear structure. Panel (d) shows the ground-truth nonlinear component computed directly from (2)–(3), confirming that the residual is a deterministic, spatially structured target rather than noise.

Figure 3 illustrates the wafer-to-wafer evolution of the per-wafer mean overlay and 3σ across all 50 wafers for each lot. The four qualitatively distinct drift regimes are clearly visible: monotonic creep in Lots 1–2, accelerating multi-parameter drift in Lot 3, and the characteristic oscillatory signature of chiller cycling in Lot 4. Polynomial thermal detrending (degree $K = 2$) removes these trends before Brink fitting, reducing the effective noise seen by the model.

The spatial structure of the nonlinear residual across all four lots is shown in Fig. 4. The pattern is fixed to the intra-field die grid, grows gently over the lot (30% amplitude increase),

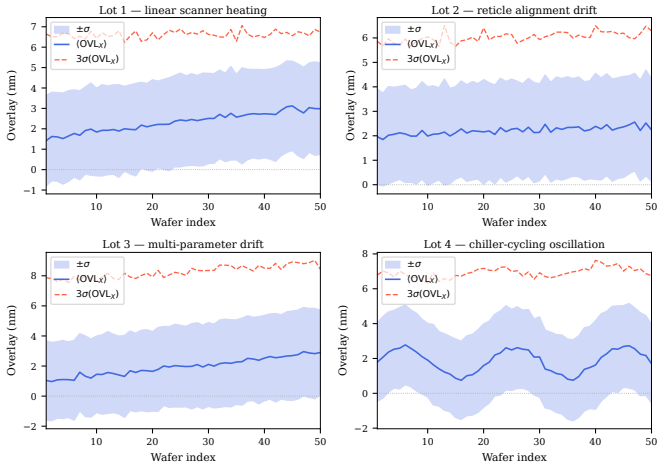

Fig. 3. Per-wafer temporal drift profiles across all 50 wafers in each lot. Blue solid: per-wafer mean OVL_X. Red dashed: per-wafer $3\sigma(\mathrm{OVL}_X)$. The four drift scenarios (linear heating, reticle drift, multi-parameter drift, chiller oscillation) are clearly distinguishable and span the primary mechanisms reported in EUV production environments [5], [6].

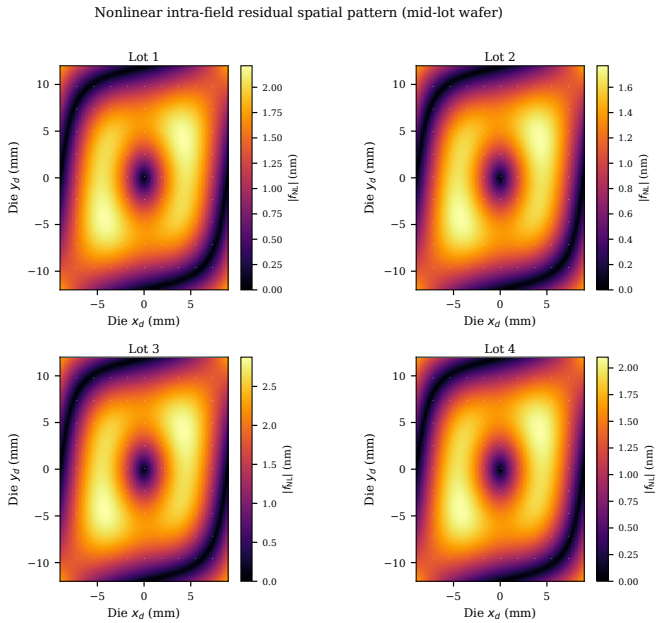

Fig. 4. Nonlinear intra-field residual f_{NL} evaluated on the die-coordinate grid for a mid-lot wafer ($w = 25$). Colour encodes magnitude $|f_{\mathrm{NL}}|$ (nm); arrows show direction. The pattern is identical for all 1 420 sites per wafer and grows by 30% from wafer 1 to wafer 50. Lot 3 has the largest amplitude ($A = 1.30$); Lot 2 the smallest ($A = 0.80$).

and shows distinct orientation and amplitude per lot. Lot 3 has the largest amplitude ($A_3 = 1.30$), consistent with its strongest Brink residual in Table II.

V. Benchmark Experiments

To validate the dataset's utility, we evaluate a two-stage correction pipeline — identical in spirit to the one introduced in [7] but operating on overlay rather than Zernike coefficients — comprising: (i) a cross-wafer Brink fit (Stage 1), and (ii) a

TABLE III
Benchmark: Brink 3σ vs. Brink+PGNN 3σ (nm)

Lot	Brink	Brink+PGNN (N_{train})			Improv.
		$n=3$	$n=5$	$n=10$	($n=10$)
1	1.782	1.024	1.012	1.005	43.6%
2	1.532	1.031	0.998	0.976	36.3%
3	2.207	1.142	1.112	1.068	51.6%
4	2.578	2.527	2.126	2.181	15.4%
Mean					**36.7%**

Physics-Guided Neural Network (PGNN) residual correction trained on the Brink residuals of $N_{\mathrm{train}} \in \{3, 5, 10\}$ training wafers [12]. The PGNN is a 3-hidden-layer MLP with $\tanh$ activations taking normalised die coordinates $(\tilde{x}, \tilde{y})$ as inputs, regularised by a spatial smoothness penalty on the gradient $\|\nabla_{(\tilde{x},\tilde{y})} f\|^2$.

Table III reports the overlay 3σ on held-out test wafers across all four lots and training set sizes. The PGNN achieves a **33–52%** reduction in Brink 3σ at $N_{\mathrm{train}} = 10$, with a 100% per-wafer win rate across all conditions. Importantly, the improvement is **near-constant** with respect to training set size: as few as **three training wafers** suffice to identify the nonlinear fingerprint, a property that directly supports real-time APC deployment at lot start.

The lower improvement on Lot 4 (15.4%) is expected: the sinusoidal chiller-cycling drift induces a wafer-to-wafer variation in the Brink parameters (T_x, T_y) that, after polynomial detrending, leaves a residual that is partially time-varying. This makes Lot 4 a harder benchmark scenario, and its distinct character from Lots 1–3 is an intentional feature of the dataset design.

VI. Reproducibility and Data Access

The complete dataset generator is the single file `src/generate_synthetic_data.py` in the companion repository. Executing

```
python src/generate_synthetic_data.py \
        --outdir dat_synthetic --seed 42
```

reproduces the 200 CSV files constituting the full dataset (`LotNWaferM.csv`, $N \in \{1, \ldots, 4\}$, $M \in \{1, \ldots, 50\}$) with bit-exact reproducibility across platforms (NumPy `default_rng`, seed 42). Each file contains 1 420 rows with columns `WaferCenterX_mm_`, `WaferCenterY_mm_`, `DieCenterX_mm_`, `DieCenterY_mm_`, `OverlayX`, `OverlayY`. The schema is identical to the industrial APC metrology export format, enabling drop-in replacement of real data in existing correction pipelines.

VII. Conclusion

We have presented EUV-OVL-SYN, a physically grounded, fully reproducible synthetic benchmark dataset for overlay correction research in EUV lithography. The dataset faithfully replicates the signal hierarchy of real scanner data — a

dominant Brink polynomial fingerprint, calibrated temporal drift in four production-relevant scenarios, and a nonlinear intra-field residual analytically disjoint from the Brink polynomial basis — while providing full ground-truth decomposition unavailable from industrial sources. A companion benchmark demonstrates that a PGNN two-stage correction pipeline achieves 33–52% reduction in overlay 3σ over the Brink-only baseline with a 100% per-wafer win rate, confirming that the dataset is both non-trivially challenging and solvable. EUV-OVL-SYN is intended as a shared substrate for the semiconductor metrology community to develop, compare, and rigorously validate the next generation of overlay correction algorithms.

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